



Unit 2 : Lec 4 (BLDC)

★ PM-BLDC :

uses High-energy PM for field excitation

Provides

↳ High power density

↳ High speed

↳ High efficiency

BLDC motor is promising motor for EV & HEV

★ Advantages :

High η # Compactness # Ease of control

Ease of cooling # low maintenance , long life .

low noise .

★ Disadvantages of BLDC Motor

High Cost

Limited Constant-Power Range

Safety

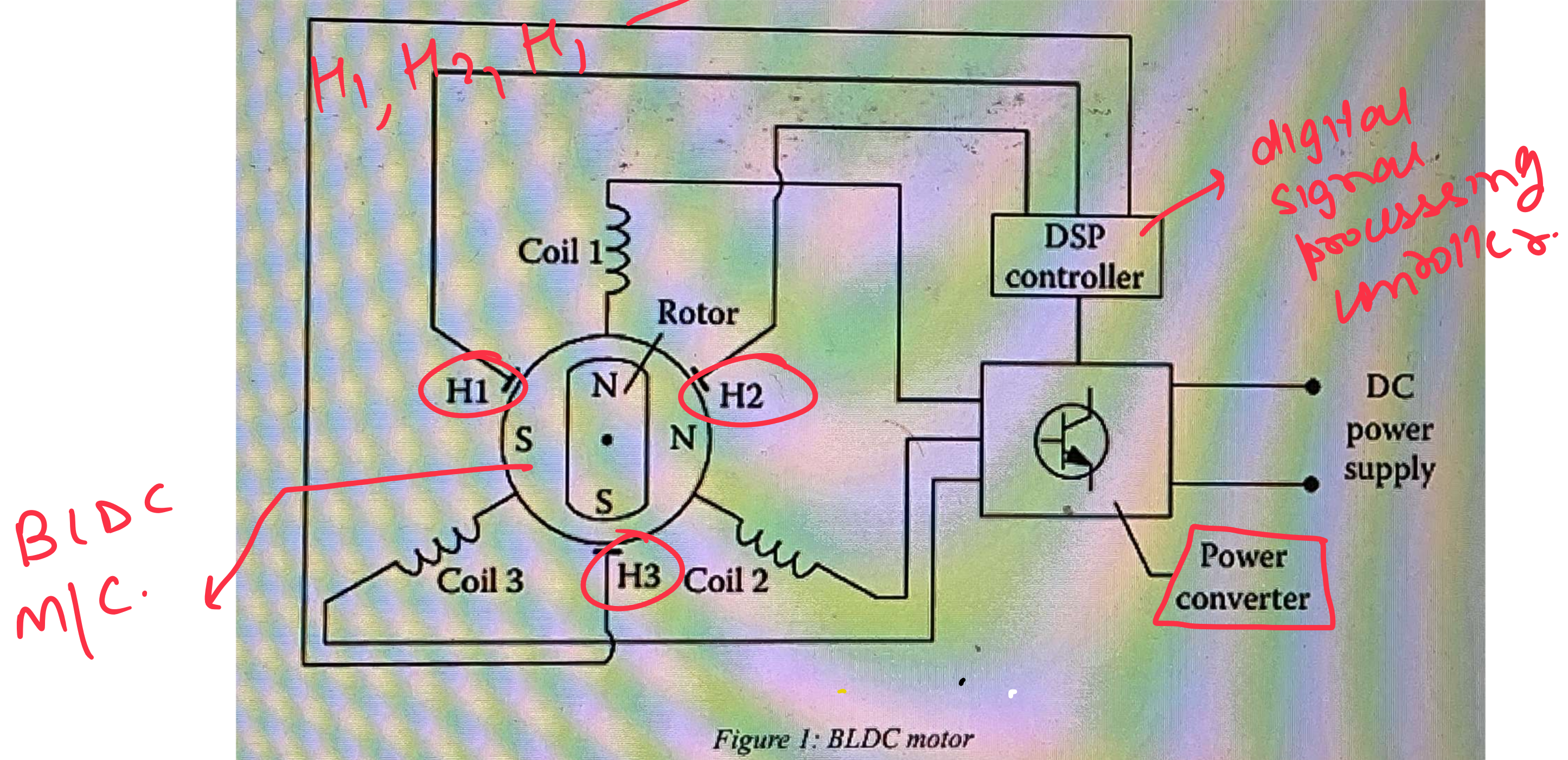
Magnet Demagnetization

High Speed Capability (x)

Inverter Failure.

★ BLDC Motor Drive: Basic Principle.

- Hall / Position
Sensor



Working:

- Position Sensors H_1, H_2, H_3 detect rotor position
- Position information goes to DSP controller
- DSP generates gating signals
- Power converter switches app. stator wdg
- Controls ω & speed.

Basic }

Position Sensor $\rightarrow$ DSP controller $\rightarrow$ PC }

Flow }

BLDC
Motor

★ BLDC Rotor Construction

① Surface Mounted PM Rotor:

- # PM's are mounted on rotor surface
- # Easy to construct
- # Skewed poles can reduce cogging torque
- # PM's can potentially fly apart at high speed

② Interior - Mounted PM Rotor :-

- # PM's are placed inside the rotor
- # less common than surface mounted
- # Suitable for high speed
- # Has inductance variation because PM region behaves approx like

air in magnetic-circuit calculations.

★ BLDC Classification Based On Back Emf :-

①

TRAPAZOIDAL BACK EMF BLDC:

Ideal Characteristics

- Rectangular distribution of magnetic flux in air gap
- Rectangular current waveform
- Concentrated stator wdg

Excitation :

- Quasi-square waveform
- Two 60° electrical intervals of zero current excitation per cycle
- only six commutation instants are required per electrical cycle.

- $\therefore$, rotor position sensor resolution requirement is relatively low.

②

3 ϕ TRAPAZOIDAL BLDC:

- Phase back emf's are displayed by 120° electrical.
- Current is of 120° ON 60° OFF type.
- Each phase conducts for 240° total
 - ↳ 120° $\rightarrow$ +ve
 - ↳ 120° $\rightarrow$ -ve.
- Current pulses should be synchronised with corresponding back EMF for maximum & constant torque per ampere

120° ON + 60° OFF

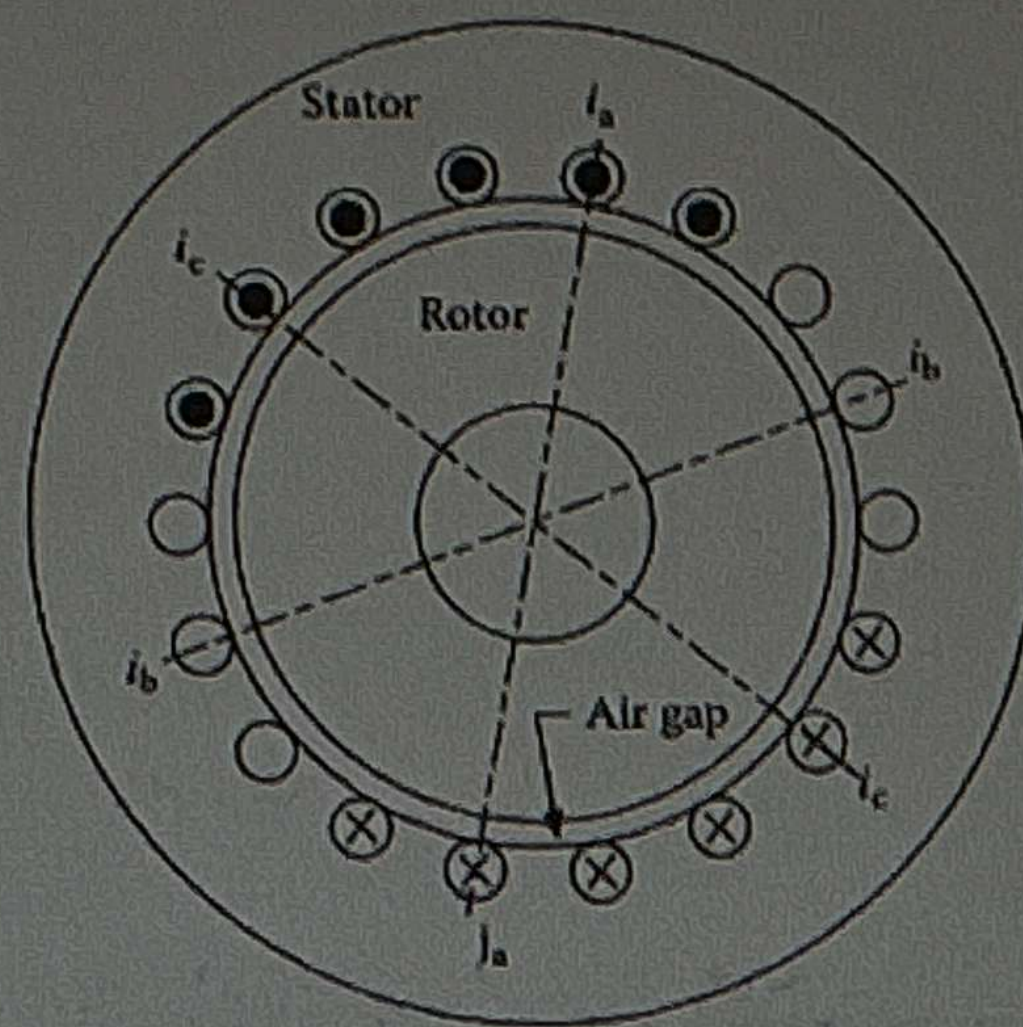
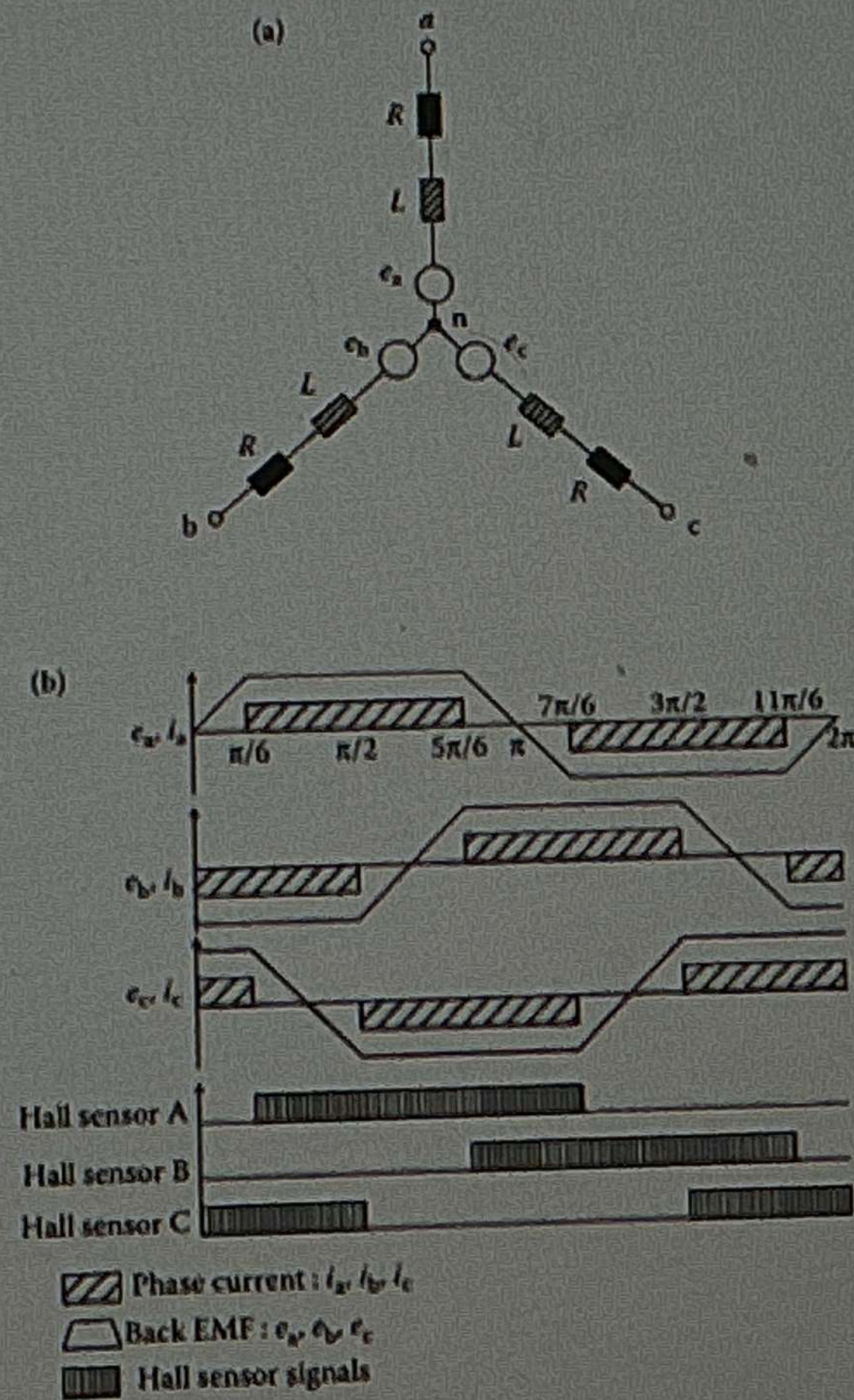


Figure 3: Winding configuration of the trapezoidal-shaped back EMF BLDC.



★ Sinusoidal Back EMF BLDC

Ideal:

- # Sinusoidal distribution of ϕ
- # Sinusoidal current waveform
- # Sinusoidal distribution of stator conductors

Operation

- # Back EMF of each phase is sinusoidal
- # Similar to an AC synchron. motor
- # Stator produces a rotating MMF wave.
- # Can be analyzed using phasor diagram

★ BLDC Performance (Basic Idea)

Torque is produced by interaction b/w

↳ $\vec{B}$

↳ Current

PM produces the magnetic field

Current depends on :

- Source V_g
- Controller
- Back emf

Back emf depends on

- $\vec{B}$
- Rotor speed

Current therefore must
be controlled to obtain
desired torque & speed

★ Assumption for BLDC

Motor is not saturated

Stator resistance are equal

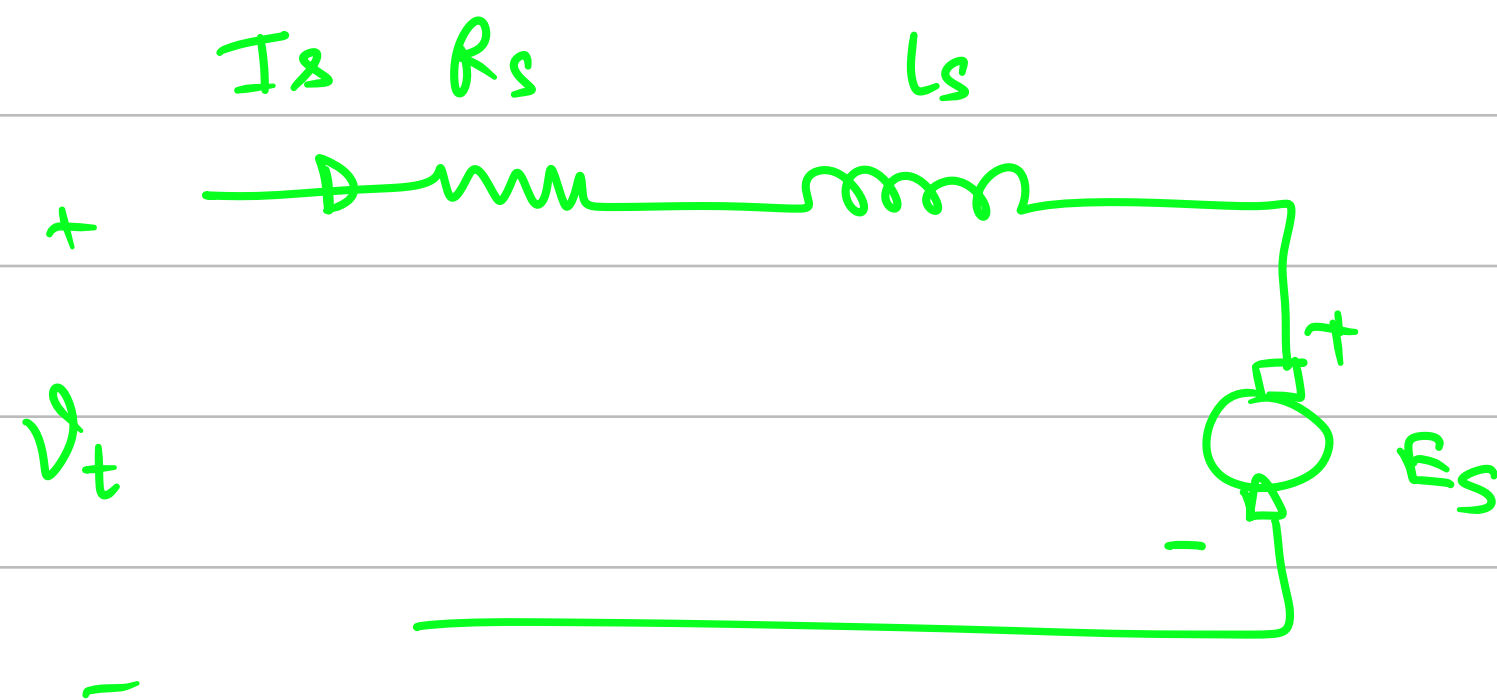
& inductances are constant

Power Semiconductor devices

are ideal

Iron losses are negligible.

★ BLDC equivalent ckt :



$V_t = \text{supply } V_g$

$R_s = \text{winding resistance}$

$L_s = \text{leakage inductance}$

$E_s = \text{Back emf}$

$$L_s = L - M$$

equations:

$$V_t = R_s I_s + L_s \frac{dI_s}{dt} + E_s$$

Back emf :

$$E_s = K_E \omega_r$$

Electro-Magnetic Torque

$$T_e = K_T I_s$$

Mechanical eqⁿ :

$$T_e = T_L + J \frac{d\omega_r}{dt} + B \omega_r$$

$K_E = B$ or any const

$k_T = \text{Torque constant}$

$\omega_r = \text{rotor angular speed}$

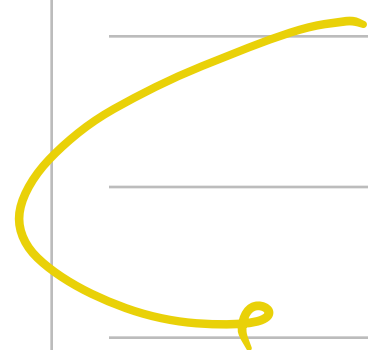
$T_L = \text{load Torque}$

$J = \text{moment of inertia}$

$B = \text{viscous resistance coefficient}$

★ Steady-State BLDC Torque Eqⁿ

$$T_e = \frac{(V_t - K_e \omega_r) k_T}{R_s}$$



As $\omega_r \uparrow$

↳ Back emf $\uparrow$

↳ $I \downarrow$

↳ $\tau \downarrow$

★ Constant V_g Operation

- At low speed, especially during starting
 - ↳ Back EMF is low
 - ↳ Current becomes very high
 - ↳ Torque becomes very high
- Excessive starting current can damage stator windings
- therefore, variable V_g control is required to limit current
- Variable V_g allows maximum constant torque

★ Dynamic / Transient Operation

Laplace / Domain Eqⁿ:

$$V_t(s) = E_s(s) + (R_s + sL_s) I_s(s)$$

$$E_s(s) = k_E \omega_r(s)$$

$$T_e(s) = k_T I_s(s)$$

$$T_e(s) = T_L(s) + (B + sJ) \omega_r(s)$$

Note

L_s = determines how quickly current builds up
 J = determines how quickly speed builds up

